WORKOUT 1

```
import statistics as st
ages = [25, 30, None, 28, 120, 35, None, 29, 31, 27]
valid = [a for a in ages if a is not None]
med = st.median(valid)
filled = [a if a is not None else med for a in ages]
q1, q2, q3 = st.quantiles(filled, n=4)
iqr = q3 - q1
lo = q1 - 1.5 * iqr
hi = q3 + 1.5 * iqr
outliers = [a for a in filled if a < lo or a > hi]
capped = [min(max(a, lo), hi) for a in filled]
mn, mx = min(capped), max(capped)
scaled = [round((a - mn) / (mx - mn), 3) for a in capped]
print("Median used:", med)
print("Imputed:", filled)
print("Fences:", (round(lo, 2), round(hi, 2)))
print("Outliers:", outliers)
print("Capped:", [round(a, 2) for a in capped])
print("Min-max scaled:", scaled)
```

```
Median used: 29.5
Imputed: [25, 30, 29.5, 28, 120, 35, 29.5, 29, 31, 27]
Fences: (21.38, 38.38)
Outliers: [120]
Capped: [25, 30, 29.5, 28, 38.38, 35, 29.5, 29, 31, 27]
Min-max scaled: [0.0, 0.374, 0.336, 0.224, 1.0, 0.748, 0.336, 0.299, 0.449, 0.15]
```

WORKOUT 3

```
print(row)
```

```
Training rows: 6
Test rows: 2

Training amount mean: 858.33
Training amount std: 385.59

Leakage check:
Mean used for scaling: 858.33
Test mean (comparison only): 810
Scaling statistics came from training rows only.

Feature correlations:
amount          r = -0.641 keep = True
items           r = +0.816 keep = True
weekday         r = -0.465 keep = False
is_weekend      r = -0.500 keep = False
amount_per_item r = -0.739 keep = True
scaled_amount   r = -0.641 keep = True

Selected features: ['amount', 'items', 'amount_per_item', 'scaled_amount']

Final feature table:
{'date': '2026-03-02', 'amount': 500, 'items': 5, 'weekday': 0, 'is_weekend': 0, 'amount_per_item': 100.0, 'target': 1, 'lag_amount': None, '
{'date': '2026-03-07', 'amount': 1200, 'items': 3, 'weekday': 5, 'is_weekend': 1, 'amount_per_item': 400.0, 'target': 0, 'lag_amount': 500, '
{'date': '2026-03-08', 'amount': 900, 'items': 4, 'weekday': 6, 'is_weekend': 1, 'amount_per_item': 225.0, 'target': 1, 'lag_amount': 1200, '
{'date': '2026-03-09', 'amount': 450, 'items': 5, 'weekday': 0, 'is_weekend': 0, 'amount_per_item': 90.0, 'target': 1, 'lag_amount': 900, 'ro
{'date': '2026-03-14', 'amount': 1500, 'items': 3, 'weekday': 5, 'is_weekend': 1, 'amount_per_item': 500.0, 'target': 0, 'lag_amount': 450, '
{'date': '2026-03-17', 'amount': 600, 'items': 6, 'weekday': 1, 'is_weekend': 0, 'amount_per_item': 100.0, 'target': 1, 'lag_amount': 1500, '
{'date': '2026-03-21', 'amount': 1100, 'items': 2, 'weekday': 5, 'is_weekend': 1, 'amount_per_item': 550.0, 'target': 0, 'lag_amount': 600, '
{'date': '2026-03-24', 'amount': 520, 'items': 4, 'weekday': 1, 'is_weekend': 0, 'amount_per_item': 130.0, 'target': 0, 'lag_amount': 1100, '
```

WORKOUT 4

```
            | , reason
        )

    print("\nTotal frequent itemsets:", len(frequent))
    print("Total pruned candidates:", len(pruned))
```

```
Frequent 1-itemsets:
['beer'] support count = 3
['bread'] support count = 4
['diaper'] support count = 4
['milk'] support count = 4

Frequent 2-itemsets:
['beer', 'diaper'] support count = 3
['bread', 'diaper'] support count = 3
['bread', 'milk'] support count = 3
['diaper', 'milk'] support count = 3

Pruned candidate itemsets:
['cola'] support count = 2 | Below minimum support
['eggs'] support count = 1 | Below minimum support
['beer', 'bread'] support count = 2 | Pruned: below minimum support
['beer', 'milk'] support count = 2 | Pruned: below minimum support
['beer', 'bread', 'diaper'] support count = 2 | Pruned: a subset is not frequent
['beer', 'diaper', 'milk'] support count = 2 | Pruned: a subset is not frequent
['bread', 'diaper', 'milk'] support count = 2 | Pruned: below minimum support

Total frequent itemsets: 8
Total pruned candidates: 7
```

WORKOUT6

```
ELBOW METHOD RESULTS
----------------------------------------

k = 1
Centroids: [(3.14, 4.07)]
SSE: 37.071

k = 2
Centroids: [(1.25, 1.5), (3.9, 5.1)]
SSE: 8.525

k = 3
Centroids: [(3.62, 4.62), (1.25, 1.5), (5.0, 7.0)]
SSE: 2.5

k = 4
Centroids: [(1.0, 1.0), (3.62, 4.62), (5.0, 7.0), (1.5, 2.0)]
SSE: 1.875

k = 5
Centroids: [(4.0, 5.0), (1.0, 1.0), (5.0, 7.0), (3.25, 4.25), (1.5, 2.0)]
SSE: 0.75

ELBOW SUMMARY
----------------------------------------
k = 1, SSE = 37.071
k = 2, SSE = 8.525
k = 3, SSE = 2.500
k = 4, SSE = 1.875
k = 5, SSE = 0.750

Compare the SSE values to identify the elbow.
```

WORKOUT 7

```
Sorted k-distance list:
[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 23.345]
Largest gap: 22.345
Suggested eps: 1.0

DBSCAN Results:
(1, 1) Cluster: 1 | core
(1, 2) Cluster: 1 | core
(2, 1) Cluster: 1 | core
(2, 2) Cluster: 1 | core
(8, 8) Cluster: 2 | core
(8, 9) Cluster: 2 | core
(9, 8) Cluster: 2 | core
(9, 9) Cluster: 2 | core
(25, 25) Cluster: -1 | noise

Clusters found: 2
```